Logical laws, normal forms, and TT-completeness

Introduction to Logic (CS), Week 4

Dan Frumin

University of Groningen

Monday 21 and Tuesday 22 Sep 2026

Some Organisational Matters

HW1: grades will be out in BS, model solutions will be there as well

Midterm Exam: next week, please check the Rooster

- Pen-and-paper, closed book
- Topics: everything up to this lecture...
- + today's lecture
- Instructions will be posted in Brightspace. Please read them
- Old midterms also in BS

Midterm topics

- Validity of arguments
- Translating from English to FOL
- Fitch proofs
- Making truth tables
- Identifying contradictions, tautologies, and consequences
- Rewriting sentences into normal forms (this week)

Today's Lecture

1. Tautologies, consequences, equivalences
2. Truth-Functional Semantics and Completeness
3. Going over exercises

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Tautologies, Contradictions and Possibilities

Sentence is ...	if main connective column in truth table has...
a *tautology*	T in all rows
a *logical truth*	T in all non-spurious rows
a *(TT-)contradiction*	F in all rows
a *logical contradiction*	F in all non-spurious rows
TT-possible	T in at least one row
logically possible	T in at least one non-spurious row

Summarizing the notions

Single proposition	Multiple propositions	
Tautology	Taut. Consequence $P \Vdash Q$	Taut. Equivalence $P \equiv Q$
Logical Truth	Log. Consequence $P \vdash Q$	Log. Equivalence $P \Leftrightarrow Q$

Summarizing the notions

Single proposition	Multiple propositions	
Tautology	Taut. Consequence $P \Vdash Q$	Taut. Equivalence $P \equiv Q$
Logical Truth	Log. Consequence $P \vdash Q$	Log. Equivalence $P \Leftrightarrow Q$

To remember those different concepts understand the axis along which the notions change.

Checking consequences, equivalence

Suppose you are asked to check whether two sentences are (tauto)logically equivalent, or whether one is the consequence of another.

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- Logical consequence:
  $P_1, \ldots, P_n \vdash Q$ iff $P_1 \wedge \cdots \wedge P_n \rightarrow Q$ is logical truth
- Tautological consequence:
  $P_1, \ldots, P_n \Vdash Q$ iff $P_1 \wedge \cdots \wedge P_n \rightarrow Q$ is tautology
- Logical equivalence:
  $P \Leftrightarrow Q$ iff $P \leftrightarrow Q$ is logical truth
- Tautological equivalence:
  $P \equiv Q$ iff $P \leftrightarrow Q$ is tautology

Examples of logical equivalences

Logical equivalence is one of the "strongest" notions of equivalence between two propositions; we will return to it in this lecture.

law of double negation	P	$\Leftrightarrow$	$\neg\neg P$
definability of $\rightarrow$	$P \rightarrow Q$	$\Leftrightarrow$	$\neg P \vee Q$
definability of $\leftrightarrow$	$P \leftrightarrow Q$	$\Leftrightarrow$	$(P \rightarrow Q) \wedge (Q \rightarrow P)$

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Meta-level vs object-level

Important:

- The symbols $\Leftrightarrow$, $\equiv$ is meta-notation; it is not a relation symbol in FOL/PL. In particular, $P \leftrightarrow Q$ is not a sentence in FOL/PL. (Meta notation vs object notation.)
- Logical equivalences cannot be used as proof steps in system $\mathcal{F}$ proofs. Formal proofs are syntactic objects. Different formulas require different rules. (The Fitch tool from the book allows you to use them via a special rule)

Another prong in the logic fork

So far we have seen various tools for analyzing sentences and arguments: truth tables, Fitch proofs

We have applied those tools to develop notions of equivalence. But those things offer us another way of looking at logic: through equations.

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- Arithmetic as equations for $+, -, *$
- Logic as equations for $\wedge, \vee, \neg$

The Algebra of Logic

Arithmetic:

$$x + y = y + x[]\ xy = yx$$
$$x(y + z) = xy + xz$$

Logic:

$$P \vee Q \Leftrightarrow Q \vee P$$
$$P \wedge Q \Leftrightarrow Q \wedge P$$
$$P \wedge (Q \vee R) \Leftrightarrow (P \wedge Q) \vee (P \wedge R)$$

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$$\neg\neg P \Leftrightarrow P$$
$$P \wedge \neg P \Leftrightarrow \bot$$
$$P \vee \bot \Leftrightarrow P$$

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Practice: verify the equivalences in the right column

Towards rewriting sentences

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$$P \leftrightarrow Q \Leftrightarrow (P \rightarrow Q) \wedge (Q \rightarrow P) \Leftrightarrow (\neg P \vee Q) \wedge (\neg Q \vee P)$$

Rewriting with Equivalences:

- Using chains of equivalences for simplifying or reducting sentences
- Often the goal is to reduce a sentence to some restricted form (useful for applications like SAT-solvers)
- What are some of these restricted forms and what are the correct principles for such rewriting?

Principles of Rewriting: Substitution

We write $S(P)$ to denote a sentence in FOL that contains the sentence P.
Example: $S(P) = \neg(P \vee (A \rightarrow (B \leftrightarrow P)))$.

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We form $S(Q)$ by **substituting** sentence Q **for** P in $S(P)$.
For example, $S(P \rightarrow R) = \neg((P \rightarrow R) \vee (A \rightarrow (B \leftrightarrow (P \rightarrow R))))$.

Where have we seen a similar notion substitution used?

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If $P \Leftrightarrow Q$, then $S(P) \Leftrightarrow S(Q)$.

In words: if P and Q are logically equivalent, then so are $S(P)$ and $S(Q)$.

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Substitution principle

If $P \Leftrightarrow Q$, then $S(P) \Leftrightarrow S(Q)$.

In words: if P and Q are logically equivalent, then so are $S(P)$ and $S(Q)$.

We already used it in our example:

$$P \leftrightarrow Q \Leftrightarrow (P \rightarrow Q) \wedge (Q \rightarrow P) \Leftrightarrow (\neg P \vee Q) \wedge (\neg Q \vee P)$$

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$$A \rightarrow (B \leftrightarrow C)$$

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More serious application: restricted negation

Negation Normal Form

A sentence is in *negation normal form* (NNF) if

- it is built from atomic sentences using $\wedge, \vee, \neg$,
- and $\neg$ is only applied to atomic sentences.

E.g. $\neg P$ is allowed, but $\neg(Q \vee R)$ is not

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We apply the same technique of substitution, using the *De Morgan laws* and *the law of double negation*:

$$\neg(P \wedge Q) \quad \Leftrightarrow \quad \neg P \vee \neg Q$$

$$\neg(P \vee Q) \quad \Leftrightarrow \quad \neg P \wedge \neg Q$$

$$\neg\neg P \quad \Leftrightarrow \quad P$$

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$\neg(A \vee B) \vee C \quad \Leftrightarrow \quad$ (De Morgan: $\neg(P \vee Q) \Leftrightarrow \neg P \wedge \neg Q$)

$(\neg A \wedge \neg B) \vee C$

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$\neg((A \vee B) \wedge \neg C)$	$\Leftrightarrow$	(De Morgan: $\neg(P \wedge Q) \Leftrightarrow \neg P \vee \neg Q$)
$\neg(A \vee B) \vee \neg\neg C$	$\Leftrightarrow$	(law of double negation: $\neg\neg P \Leftrightarrow P$)
$\neg(A \vee B) \vee C$	$\Leftrightarrow$	(De Morgan: $\neg(P \vee Q) \Leftrightarrow \neg P \wedge \neg Q$)
$(\neg A \wedge \neg B) \vee C$	is	in Negation Normal Form

Logical Laws in Normal Forms

We can rewrite with any logical equivalence, in principle. In practice, for rewriting into NNF we used various *logical laws* — interesting and useful logical equivalences:

law of double negation	$\neg\neg P$	$\Leftrightarrow$	P
De Morgan	$\neg(P \wedge Q)$	$\Leftrightarrow$	$(\neg P \vee \neg Q)$
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definability of $\leftrightarrow$	$P \leftrightarrow Q$	$\Leftrightarrow$	$(P \rightarrow Q) \wedge (Q \rightarrow P)$

Logical Laws for $\wedge$ and $\vee$

idempotence of $\wedge$	$P \wedge P$	$\Leftrightarrow$	P
idempotence of $\vee$	$P \vee P$	$\Leftrightarrow$	P
commutativity of $\wedge$	$P \wedge Q$	$\Leftrightarrow$	$Q \wedge P$
commutativity of $\vee$	$P \vee Q$	$\Leftrightarrow$	$Q \vee P$
associativity of $\wedge$	$(P \wedge Q) \wedge R$	$\Leftrightarrow$	$P \wedge (Q \wedge R)$
associativity of $\vee$	$(P \vee Q) \vee R$	$\Leftrightarrow$	$P \vee (Q \vee R)$

We call these the ACI laws (abbreviation of Associativity, Commutativity, Idempotence)

Example: Associativity, Commutativity, Idempotence

$((C \wedge A) \wedge (B \wedge (C \wedge B))) \wedge C$

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$$(C \wedge A) \wedge (B \wedge C) \quad \Leftrightarrow \quad \text{(commutativity of } \wedge\text{, twice)}$$

$$(A \wedge C) \wedge (C \wedge B)$$

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$A \wedge (C \wedge (C \wedge B)) \quad \Leftrightarrow \quad$ (associativity,idempotence of $\wedge$)

$A \wedge (C \wedge B)$

Example: Associativity, Commutativity, Idempotence

$$((C \wedge A) \wedge (B \wedge (C \wedge B))) \wedge C \quad \Leftrightarrow \quad \text{(commutativity of } \wedge\text{)}$$

$$((C \wedge A) \wedge (B \wedge (B \wedge C))) \wedge C \quad \Leftrightarrow \quad \text{(associativity of } \wedge\text{)}$$

$$((C \wedge A) \wedge ((B \wedge B) \wedge C)) \wedge C \quad \Leftrightarrow \quad \text{(idempotence of } \wedge\text{)}$$

$$((C \wedge A) \wedge (B \wedge C)) \wedge C \quad \Leftrightarrow \quad \text{(associativity of } \wedge\text{)}$$

$$(C \wedge A) \wedge ((B \wedge C) \wedge C) \quad \Leftrightarrow \quad \text{(associativity, idempotence of } \wedge\text{)}$$

$$(C \wedge A) \wedge (B \wedge C) \quad \Leftrightarrow \quad \text{(commutativity of } \wedge\text{, twice)}$$

$$(A \wedge C) \wedge (C \wedge B) \quad \Leftrightarrow \quad \text{(associativity of } \wedge\text{)}$$

$$A \wedge (C \wedge (C \wedge B)) \quad \Leftrightarrow \quad \text{(associativity,idempotence of } \wedge\text{)}$$

$$A \wedge (C \wedge B) \quad \Leftrightarrow \quad \text{(commutativity of } \wedge\text{)}$$

$$A \wedge (B \wedge C)$$

Associativity, Commutativity, Idempotence (ACI laws)

In general:

- ACI laws of $\wedge$ allow you to ignore parentheses, reorder and remove duplicates.

$$((C \wedge A) \wedge (B \wedge (C \wedge B))) \wedge C \overset{ACI}{\Leftrightarrow} A \wedge (B \wedge C)$$

- Similarly for $\vee$.
- But not when $\wedge$ and $\vee$ both occur!

In the midterm/exam, when you prove logical equivalences using substitution, you need to give the detailed steps as in the example on the previous slide.

Annotated Chain-of-Equivalences Proofs

General form of proof of $A \Leftrightarrow C$ using substitution principle:

$A \quad \Leftrightarrow \quad$ (reason why $A \Leftrightarrow B_1$)

$B_1 \quad \Leftrightarrow \quad$ (reason why $B_1 \Leftrightarrow B_2$)

$B_2 \quad \Leftrightarrow \quad$ (reason why $B_2 \Leftrightarrow B_3$)

$\vdots$

$B_n \quad \Leftrightarrow \quad$ (reason why $B_n \Leftrightarrow C$)

C

Disjunctive Normal Form

Recall: Negation normal form (NNF)

A sentence S is in NNF if $\neg$ occurs only in front of atomic sentences.

A *literal* is an atomic sentence or a negation of an atomic sentence.
I.e., of the form P or $\neg P$ for some atomic P.

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E.g. $\neg P \vee Q$ is allowed, $P \wedge (Q \vee R)$ is not

DNF

Disjunctive Normal Form (DNF)

A sentence S is in DNF if S is a disjunction of (one or more) conjunctions of (one or more) literals.
(NNF, no disjunctions below conjunctions)

Which of the following sentences are in DNF?

- $(A \wedge \neg B \wedge C) \vee (\neg A \wedge C) \vee (B \wedge C)$
- $A \vee \neg B \vee C$
- $\neg A \wedge B$
- $\neg\neg A$
- $\neg(A \wedge B)$
- $\neg(A \vee B)$
- $A \vee (B \wedge (C \vee D))$

Rewriting to DNF

Consider, e.g. $A \vee (B \wedge (C \vee D))$. How to get it to DNF?

Rewriting to DNF

Consider, e.g. $A \vee (B \wedge (C \vee D))$. How to get it to DNF?

$$A \vee (B \wedge (C \vee D)) \Leftrightarrow \qquad \text{(left-D}\wedge\text{)}$$
$$A \vee ((B \wedge C) \vee (B \wedge D))$$

Using the distributivity laws:

Distributivity from the left:

left-D$\vee$ $\quad P \vee (Q \wedge R) \quad \Leftrightarrow \quad (P \vee Q) \wedge (P \vee R)$

left-D$\wedge$ $\quad P \wedge (Q \vee R) \quad \Leftrightarrow \quad (P \wedge Q) \vee (P \wedge R)$

Distributivity from the right:

right-D$\vee$ $\quad (Q \wedge R) \vee P \quad \Leftrightarrow \quad (Q \vee P) \wedge (R \vee P)$

right-D$\wedge$ $\quad (Q \vee R) \wedge P \quad \Leftrightarrow \quad (Q \wedge P) \vee (R \wedge P)$

Example: Rewriting to DNF

– and also simplify using ACI.

$$\neg(P \to \neg Q) \wedge (R \to Q)$$

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Example: Rewriting to DNF

– and also simplify using ACI.

$$\begin{aligned} \neg(P \to \neg Q) \wedge (R \to Q) &\Leftrightarrow (\text{Def} \to) \\ \neg(\neg P \vee \neg Q) \wedge (\neg R \vee Q) &\Leftrightarrow (\text{De Morgan}) \\ (\neg\neg P \wedge \neg\neg Q) \wedge (\neg R \vee Q) & \end{aligned}$$

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(\neg\neg P \wedge \neg\neg Q) \wedge (\neg R \vee Q) &\Leftrightarrow \text{(double-neg)} \\
(P \wedge Q) \wedge (\neg R \vee Q)
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(P \wedge Q) \wedge (\neg R \vee Q) &\Leftrightarrow (\text{left-D}\wedge) \\
((P \wedge Q) \wedge \neg R) \vee ((P \wedge Q) \wedge Q)
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(P \wedge Q) \wedge (\neg R \vee Q) &\Leftrightarrow \text{(left-D}\wedge\text{)} \\
((P \wedge Q) \wedge \neg R) \vee ((P \wedge Q) \wedge Q) &\Leftrightarrow \text{(A}\wedge\text{)} \\
((P \wedge Q) \wedge \neg R) \vee (P \wedge (Q \wedge Q)) &
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((P \wedge Q) \wedge \neg R) \vee ((P \wedge Q) \wedge Q) &\Leftrightarrow (\text{A}\wedge) \\
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((P \wedge Q) \wedge \neg R) \vee ((P \wedge Q) \wedge Q) &\Leftrightarrow (\text{A}\wedge) \\
((P \wedge Q) \wedge \neg R) \vee (P \wedge (Q \wedge Q)) &\Leftrightarrow (\text{I}\wedge) \\
((P \wedge Q) \wedge \neg R) \vee (P \wedge Q)
\end{aligned}$$

Conjunctive Normal Form

Disjunctive Normal Form (DNF)

A sentence S is in DNF if S is a disjunction of (one or more) conjunctions of (one or more) literals.
(NNF, no disjunctions below conjunctions)

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(NNF, no disjunctions below conjunctions)

Conjunctive Normal Form (CNF)

A sentence S is in CNF if S is a conjunction of (one or more) disjunctions of (one or more) literals.
(NNF, no conjunctions below disjunctions)

$\neg P \wedge (Q \vee R)$ is in CNF, but $(P \wedge Q) \vee (R \wedge S)$ is not.

Rewriting Using Logical Laws

Remember:

- Rewriting with logical equivalences is not allowed in formal proofs! Only Intro, Elim, Reit rules from system $\mathcal{F}$ are allowed.
- Formal proofs in system $\mathcal{F}$ can be used to justify a logical equivalence, but not the other way around.

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Homework for the exam:

- Make a cheat sheet with all the logical laws we went over
- What are the symmetries in the laws that you see?
- In which situations will you use which laws?
- Do not bring the cheat sheet with you to the exam

Today's Lecture

1 Tautologies, consequences, equivalences

2 Truth-Functional Semantics and Completeness

3 Going over exercises

Truth-functional completeness (LPL 7.4)

- The connectives $\neg, \vee, \wedge \rightarrow$ and $\leftrightarrow$ can be seen as truth value functions (or Boolean functions): you put in one or two truth values, and the result is a truth value. Let $\mathit{Bool} = \{T, F\}$

$$\begin{array}{lll} \bot : & & \mathit{Bool} \\ \neg : & \mathit{Bool} & \rightarrow \mathit{Bool} \\ \wedge : & \mathit{Bool}^2 & \rightarrow \mathit{Bool} \\ \vee : & \mathit{Bool}^2 & \rightarrow \mathit{Bool} \end{array}$$

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- The truth table of the connective specifies the truth value function.
- More generally, the truth table of a sentence $S(P_1, \ldots, P_n)$ with atoms $P_1, \ldots, P_n$ specifies an n-ary truth value function:

$$S : \mathit{Bool}^2 \rightarrow \mathit{Bool}$$

How many connectives

- How many (abstract) binary connectives are there?
- In other words: how many binary truth value functions $f : Bool^2 \to Bool$ are there?

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Every binary connective ($*$, say) has 4 rows in a truth table:

P	Q	$P * Q$
T	T	?
T	F	?
F	T	?
F	F	?

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P	Q	$P * Q$
T	T	?
T	F	?
F	T	?
F	F	?

In every row there are 2 choices: T or F.
So there are $2 \cdot 2 \cdot 2 \cdot 2 = 2^4 = 16$ different binary connectives.

How many connectives

In general, for there are *a lot* of possible n-ary connectives: 2^{2^n}.

$$2^{2^1} = 4 \qquad 2^{2^4} = 65536$$
$$2^{2^2} = 16 \qquad 2^{2^5} = 4294967296$$
$$2^{2^3} = 256 \qquad \ldots$$

Do we need all of them?

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Do we need all of them?
We can get away with only a few connectives
As an example: NNF, DNF/CNF

Definability of connectives: xor

- We have already seen that $\rightarrow$ and $\leftrightarrow$ can be defined in terms of $\neg, \wedge, \vee$.
- Let us consider some other connectives.
- Example: xor , the exclusive disjunction.

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T	F	T
F	T	T
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F	T	T
F	F	F

- xor is definable: we have P xor $Q \Leftrightarrow (P \vee Q) \wedge \neg(P \wedge Q)$

Binary connectives nor , nand

Two other binary connectives: nor and nand .

Binary connectives nor , nand

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P	Q	P nor Q	P nand Q
T	T	F	F
T	F	F	T
F	T	F	T
F	F	T	T

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Two other binary connectives: nor and nand .

P	Q	P nor Q	P nand Q
T	T	F	F
T	F	F	T
F	T	F	T
F	F	T	T

They are definable, too: we have

$$P \text{ nor } Q \Leftrightarrow \neg(P \vee Q) \qquad P \text{ nand } Q \Leftrightarrow \neg(P \wedge Q)$$

Definability in general

Claim: Every n-arry connectives defined by a truth table can be equally defined by a sentence $S(P_1, \ldots, P_n)$ that uses only $\neg$, $\vee$, $\wedge$.

That is, $\neg$, $\vee$ and $\wedge$ is a *complete* set of connectives.

Definability of connectives (by example)

We consider an abstract ternary connective $\clubsuit$: $Bool^3 \to Bool$.

P	Q	R	$\clubsuit(P, Q, R)$
T	T	T	T
T	T	F	T
T	F	T	F
T	F	F	F
F	T	T	T
F	T	F	F
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We want to find a sentence $S(P, Q, R)$ such that $\clubsuit(P, Q, R) \Leftrightarrow S(P, Q, R)$. **How?**

Definability of connectives (by example)

P	Q	R	$\clubsuit(P, Q, R)$
T	T	T	T
T	T	F	T
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T	F	F	F
F	T	T	T
F	T	F	F
F	F	T	T
F	F	F	F

The truth table says: $\clubsuit(P, Q, R)$ is true iff

- $P = T$, $Q = T$ and $R = T$, or
- $P = T$, $Q = T$ and $R = F$, or
- $P = F$, $Q = T$ and $R = T$, or
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Idea: Encode these truth conditions in a formula.

Definability of connectives (by example)

P	Q	R	$\clubsuit(P,Q,R)$	
T	T	T	T	$(P \wedge Q \wedge R) \vee$
T	T	F	T	$(P \wedge Q \wedge \neg R) \vee$
T	F	T	F	
T	F	F	F	
F	T	T	T	$(\neg P \wedge Q \wedge R) \vee$
F	T	F	F	
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F	F	F	F	

We get a formula:

$$\clubsuit(P,Q,R) \Leftrightarrow (P \wedge Q \wedge R) \vee (P \wedge Q \wedge \neg R) \vee (\neg P \wedge Q \wedge R) \vee (\neg P \wedge \neg Q \wedge R)$$

Definability of connectives (by example)

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And it is in DNF!

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Can be simplified to: $\clubsuit(P,Q,R) \Leftrightarrow (P \wedge Q) \vee (\neg P \wedge R)$

We get a formula:

$$\clubsuit(P,Q,R) \Leftrightarrow (P \wedge Q \wedge R) \vee (P \wedge Q \wedge \neg R) \vee (\neg P \wedge Q \wedge R) \vee (\neg P \wedge \neg Q \wedge R)$$

And it is in DNF!

Truth-functional completeness

What we have shown is truth-functional completeness for $\neg, \wedge, \vee$. I.e. *the set* $\{\neg, \wedge, \vee\}$ (of Boolean connectives) *is truth-functionally complete*

Can we do it with two connectives?

Truth-functional completeness

What we have shown is truth-functional completeness for $\neg, \wedge, \vee$. I.e. *the set* $\{\neg, \wedge, \vee\}$ (of Boolean connectives) *is truth-functionally complete*

Can we do it with two connectives? *Yes*, with $\{\wedge, \neg\}$ and $\{\vee, \neg\}$:

$$P \vee Q \Leftrightarrow \neg\neg(P \vee Q) \Leftrightarrow \neg(\neg P \wedge \neg Q),$$

and similarly

$$P \wedge Q \Leftrightarrow \neg(\neg P \vee \neg Q).$$

Since we can express $\vee$ in terms of $\wedge, \neg$ (and the other way around with $\wedge$), we can conclude that $\{\wedge, \neg\}$ is also functionally complete.

Truth-functional completeness

What about $\{\wedge, \vee\}$?

Truth-functional completeness

What about $\{\wedge, \vee\}$? The answer is *no*, it is not functionally complete.

But how to show it?

We want to prove a negation: the set $\{\wedge, \vee\}$ is not truth-functionally complete. How to proceed?

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But how to show it?

We want to prove a negation: the set $\{\wedge, \vee\}$ is not truth-functionally complete. How to proceed?
Hint: consider the negation $\neg : \mathit{Bool} \to \mathit{Bool}$. How to express it with $\wedge, \vee$?

Truth-functional completeness and monotonicity

We can order the truth-values $\{\mathsf{F}, \mathsf{T}\}$ as $\mathsf{F} \leq \mathsf{T}$ and $\mathsf{T} \not\leq \mathsf{F}$.
(think of $\mathsf{T} = 1$, $\mathsf{F} = 0$)
What can we say about $P \wedge \mathsf{F}$ vs $P \wedge \mathsf{T}$ independently of P?

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(think of $\mathsf{T} = 1$, $\mathsf{F} = 0$)
What can we say about $P \wedge \mathsf{F}$ vs $P \wedge \mathsf{T}$ independently of P?

$P \wedge \mathsf{F} \leq P \wedge \mathsf{T}$ and $P \vee \mathsf{F} \leq P \vee \mathsf{T}$

In other words: $\wedge$ and $\vee$ are monotone.
But negation is not!

$$P \wedge Q \not\Rightarrow \neg Q$$

Truth-functional completeness

We saw truth-functionally complete sets of two functions, but we can even do it with one function!

- With $\neg$ alone we cannot get very far...
- We have seen that $\wedge$ or $\vee$ alone cannot do much...

Truth-functional completeness

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- We have seen that $\wedge$ or $\vee$ alone cannot do much...
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  - $\neg P$ nand $\neg Q \Leftrightarrow \neg(\neg P \wedge \neg Q) \Leftrightarrow P \vee Q$

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- The same holds for nor .

Truth-functional completeness and NAND

The connective nor is truth-functionally complete, and so is nand .
So we can write every sentence using only nor (or nand).
But is that a good idea?

- NO! The sentences get very large
- NO! It is almost impossible to understand them just be reading
- But.. important in hardware design

Today's Lecture

1 Tautologies, consequences, equivalences

2 Truth-Functional Semantics and Completeness

3 Going over exercises

Validity and soundness

What is the difference between validity and soundness?

Is the following argument valid?

Femke collects all dinosaurs.
Some birds have wings.
All dinosaurs are birds.

⊢ *Femke collects some winged birds.*

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Consider the following situation. There is a winged bird, who we call Bob, and Bob is not a dinosaur. This makes the second premise true. There is a dinosaur Dino who is also a bird, but Dino has no wings. Femke's collection consists of Dino, and nothing else. Then the first and third premise are true since Dino is the only dinosaur, but since Dino has no wings, the conclusion is false.

Translation to PL

Translate the following sentence into *propositional logic*, where atomic sentences are represented by upper-case letters. Do not forget to provide the translation key.

If the name of the file is missing, the program reads from the standard input, otherwise the program reads the file until the end.

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$(F \to S) \wedge (\neg F \to E)$ or $(\neg F \vee S) \wedge (F \vee E)$ where

F : The name of the file is missing.
S : The program reads from the standard input.
E : The program reads the file until the end.

Translation to PL

Andrei is always on time, but Bianca is on time unless she misses the bus.

Translation to PL

Andrei is always on time, but Bianca is on time unless she misses the bus.
Andrei is always on time, and (if Bianca does not miss the bus, then she is on time).

$$A \wedge (\neg M \rightarrow B)$$

A : Andrei is always on time.
B : Bianca is on time.
M : Bianca misses the bus.

Translate to FOL

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(The domain of discourse consists of all people.)

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constant m:	Maria
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binary predicate $T(x, y)$:	x tutors y
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binary predicate $T(x, y)$:	x tutors y
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$$\neg(T(m,a) \vee S(a)) \rightarrow (T(p,m) \wedge T(p,a))$$

Fitch Proofs (1)

$P \to A$
$Q \to A$
$A \to B$

$(P \vee Q) \to (A \wedge B)$

Fitch Proofs (2)

$$\begin{array}{|l} a = b \\ \hline (P(a) \leftrightarrow P(b)) \vee \neg(a = b) \end{array}$$

Wrapping up

Learning goals: At the end of this week, you are able to

- use truth tables to determine tautologies, logical truth, consequence and equivalence.
- apply logical laws to compute NNF, DNF, CNF.
- express Boolean functions using propositional connectives.

Next week (Week 5):

- Midterm: Monday (material: Weeks 1-4).
- Quantifiers “forall” ($\forall$) and “exists” ($\exists$). The full power of FOL.